

The most efficient and accurate ways of comparing the accuracy and performance of embeddings are to test their performances in real applications. Hence, in order to compare the performance of Word2Vec and Doc2Vec, we will use two ways to evaluate their performances in the K-means clustering task for clustering Reddit posts.

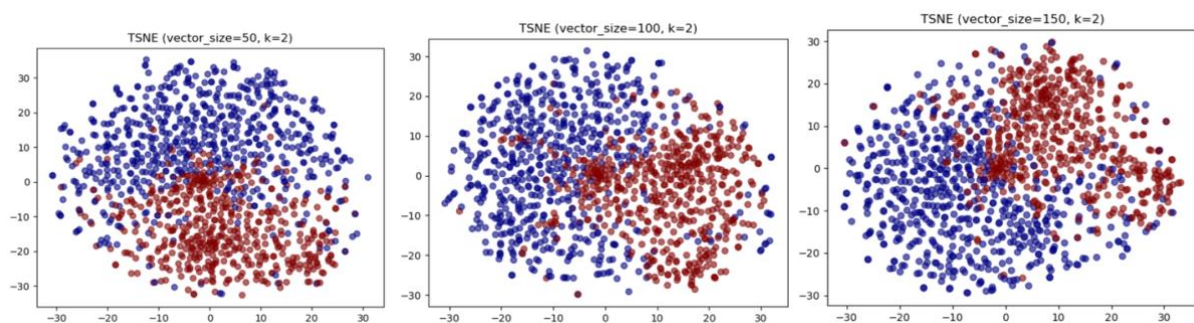
As the results, K-means clustering using Word2Vec and Doc2Vec with dimensions 50, 100, and 150 are implemented, respectively. The first evaluation method for comparing K-means clustering is silhouette scores, which examines how closely samples are inside the cluster and far samples are across clusters. The closer to 1, the better the clustering results is. The table below displays the silhouette scores of K-means clustering using these two embeddings.

Dimension	Word2Vec	Doc2Vec
50	0.3308	0.1195
100	0.2796	0.1244
150	0.2247	0.124

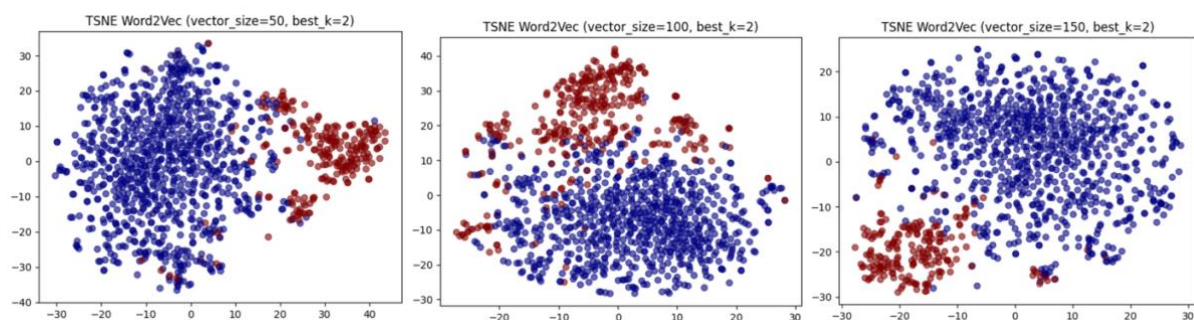
This table shows that Word2Vec outperforms Doc2Vec across all vector dimensions since the silhouette scores using Word2Vec are closer to 1 than using Doc2Vec.

Moreover, the clustering visualizations also confirm this. The visualization map of clustering results is a more intuitive and direct way of displaying the clustering performance. Figures below are K-means clustering results using these two embeddings.

Clustering visualization using Doc2Vec



Clustering visualization using Word2Vec



These figures display that the clustering using Word2Vec are mostly clear spatial separation between clusters, while Doc2Vec suffers from more overlap between clusters at all embedding dimensions. Overall, within each embedding, Word2Vec performs best at 50 dimensions with the highest silhouette score and cleanest boundaries. Doc2Vec peaks at 100 dimensions but still shows mixed clustering results. However, both embeddings degrade the clustering results at dimension 150, which might because of the curse of dimensionality. Additionally, at dimension 150, Doc2Vec maintains some balance with increased overlap, but Word2Vec collapses into one dominant cluster.

In conclusion, the Word2Vec model with 50-dimensional embeddings provides the best clustering for this dataset. This might because Doc2Vec learns embeddings by using doc vector to predict all words in the whole post, so it is more likely to capture posts' overall semantic meaning. However, Word2Vec learns embeddings based on words' frequencies and by predicting each word independently using its neighboring words, which preserves local semantic meaning and details. All Reddit posts we clustered are from the same trip topic, so they have similar overall meanings. To cluster similar posts into more cohesive and smaller clusters such as clusters for trip planning and clusters for trip experience, the difference of local details among posts are more important and useful in this case. Moreover, most of reddit posts are short, which cannot provide enough information for Doc2Vec to learn accurate semantic meanings. Hence, Word2Vec has expected better performance in this Reddit trip post clustering.